

Dawit Bonga

San Francisco, CA | U.S. Citizen | dawitbonga0@gmail.com | (415) 871-9014 | github.com/dawit-bonga | linkedin.com/in/dawit-bonga

EDUCATION

Yale University

New Haven, CT

B.S. in Computer Science & Minor in Data Science | GPA 3.71/4.0

Expected Graduation: May 2028

Relevant Coursework: Systems Programming and Computer Organization, Data Structures, Probability Theory, Linear Algebra

EXPERIENCE

Yale Interactive Machines Group

New Haven, CT

Undergraduate Research Assistant - *Robotics (C++, Python, ROS)*

Nov 2025 - Present

- Selected for an NSF-funded CAREER project deploying a public-facing social robot at the Yale Peabody Museum
- Designed and implemented a ROS2-based ZED camera pipeline enabling real-time proximity tracking for live interactions
- Led hardware upgrades for robot systems, improving stability and readiness for public museum deployment
- Built ROS2 testing and data recording pipelines, ensuring reliable performance under real-world deployment conditions

Google

New York, NY

Software Engineer Fellow - DSA, System Design

Feb 2026 - Present

- 1 of 200 admitted to 4-month fellowship providing 1:1 mentorship with Google Engineers through DSA & System Design
- Solved 40+ advanced DSA problems and mock interviews, strengthening algorithmic problem-solving and communication

Amazon

Santa Clara, CA

Software Engineer Intern - *Bedrock Finetuning infrastructure (Python/Java, Typescript,, HuggingFace)*

May 2025 - Aug 2025

- Built ML model loading pipeline via 4-bit quantization, cutting memory usage 60% and accelerating validation workflows.
- Implemented CI/CD integration tests in java for Bedrock model workflows, catching pre-production deployment issues
- Migrated validation workloads to AWS Fargate, expanding test coverage and reducing deployment-blocking issues by 30%

Yale University

New Haven, CT

Student Technician - *Systems troubleshooting (Linux, SSD, Windows)*

Oct 2024 - Present

- Resolved 25+ weekly issues across Windows, macOS, and Linux, supporting 12K+ users with hardware and OS diagnostics

TECHNICAL PROJECTS

Machine Learning Inference Engine - C++, Machine Learning

Feb 2026 - Mar 2026

- Built a C++17 neural network inference engine matching PyTorch accuracy with less than 3e-8 numerical error
- Optimized core matrix operations via cache-friendly memory access patterns, achieving a ~10x speedup (28.7 GFLOPS)
- Implemented custom tensor structure with stride-based indexing and optimized memory layout for efficient computation

Key-Value Storage Engine - C++, Storage Engineering, Database

Mar 2026 - Present

- Built a crash-safe C++ key-value store with a write-ahead log (WAL) ensuring full state recovery on restart via log replay
- Designed an append-only WAL with strict input validation and an SSTable flush layer as the foundation of an LSM storage

NBA Prop Engine - Python, Scikit-learn, Pandas, FastAPI

Oct 2024 - Dec 2024

- Trained an XGBoost model on 40K+ game logs to predict player performance, hitting a 53% prop bet win rate
- Automated data collection with a Selenium scraper and engineered 38+ features, including rolling averages and matchup stats

Mini Shell - C, POSIX APIs, Process Management:

Oct 2024 - Dec 2024

- Implemented a Unix-style shell in C supporting pipes, job control, background execution, and built-in commands.
- Engineered POSIX job control with a deferred SIGCHLD handler, ensuring reliable fg/bg/stop process state tracking.

College Express - React.js, Python, FastAPI, PostgreSQL

Jun 2025 - Mar 2026

- Built a hackathon-winning full-stack platform for AI-powered college guidance for first-gen students, reaching 40+ DAU
- Shipped user authentication, scholarship filtering, and persistent storage via Supabase and PostgreSQL

SKILLS

Languages: Python, C, C++, HTML, CSS, TypeScript, Rust

Technologies: React.js, React Native, FastAPI, HuggingFace, Scikit-learn, Pandas, XGBoost

Infrastructure & Tools: Docker, PostgreSQL, Git, ROS2, AWS (Lambda, S3, EC2, ECS, Fargate, EMR)

ADDITIONAL

Activities and Organizations: National Society of Black Engineers (NSBE), ColorStack, CodePath, Yale Prison Project

Affiliations: Gates Scholar (700 / 50,000 applicants), D.E. Shaw Connect (1/150), Google SWE Fellow (1/200),

Hack-the-Hood Winner, MLT Career prep Fellow, Questbridge Scholar,